

Research Monographaskdfjhjk

Book Author

Month Year

Mục lục

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Preface

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Phần I

First half of the book

Chương 1

Introduction

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1.1 Research question

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1.2 Answer to the question

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1.3 Positioning in the literature

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1.4 A section

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1.4.1 A subsection

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1.4.2 Yet another subsection

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1.5 A final introduction section

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Chương 2

Sections and content

Chương 3

Math

Phần II

Second half of the book

Chương 4

Cross-references and quotes

Chương 5

Thiết kế vật lý

5.1 Phân hệ Quản trị, Kiểm duyệt & Minh chứng AI

5.1.1 Cú pháp Rút gọn (Logical Schema Syntax)

- `templates` (`template_id` UUID PK, `workspace_id` UUID FK, `name` VARCHAR(150), `description` TEXT, `content_structure` TEXT, `creator_type` VARCHAR(20), `status` VARCHAR(20), `created_at` TIMESTAMP, `updated_at` TIMESTAMP)
- `template_sections` (`section_id` UUID PK, `template_id` UUID FK, `heading_title` VARCHAR(255), `order_index` INTEGER, `prompt_instruction` TEXT, `created_at` TIMESTAMP, `updated_at` TIMESTAMP)
- `placeholders` (`placeholder_id` UUID PK, `section_id` UUID FK, `variable_name` VARCHAR(100), `expected_data_type` VARCHAR(30), `description` TEXT, `is_required` BOOLEAN, `created_at` TIMESTAMP, `updated_at` TIMESTAMP)
- `template_mappings` (`mapping_id` UUID PK, `placeholder_id` UUID FK, `code_entity_id` UUID FK, `extraction_rule` TEXT, `is_active` BOOLEAN)
- `documents` (`document_id` UUID PK, `workspace_id` UUID FK, `author_user_id` UUID FK, `title` VARCHAR(255), `document_type` VARCHAR(30),

status VARCHAR(20), created_at TIMESTAMP, updated_at TIMESTAMP, deleted_at TIMESTAMP)

- document_versions (version_id UUID PK, document_id UUID FK, version_number VARCHAR(20), content TEXT, change_summary TEXT, status VARCHAR(20), ticket_codes VARCHAR(500), created_at TIMESTAMP, updated_at TIMESTAMP)
- change_logs (log_id UUID PK, version_id UUID FK, change_type VARCHAR(20), change_description TEXT, diff_content TEXT, created_at TIMESTAMP, updated_at TIMESTAMP)

5.1.2 Mã DDL SQL Hoàn Chỉnh (schema_member_4.sql)

```
CREATE TABLE templates (
    template_id          UUID                NOT NULL
        DEFAULT gen_random_uuid() ,
    workspace_id         UUID                NOT NULL,
    name                 VARCHAR(150)        NOT NULL,
    description          TEXT                NULL,
    content_structure    TEXT                NULL,
    creator_type         VARCHAR(20)         NOT NULL,
    status               VARCHAR(20)         NOT NULL
        DEFAULT 'DRAFT' ,
    created_at           TIMESTAMP           NOT NULL
        DEFAULT CURRENT_TIMESTAMP,
    updated_at          TIMESTAMP           NOT NULL
        DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_templates PRIMARY KEY (template_id) ,

    CONSTRAINT uq_templates_workspace_name UNIQUE
        (workspace_id , name) ,
    CONSTRAINT fk_templates_workspace
        FOREIGN KEY (workspace_id) REFERENCES
            workspaces (workspace_id)
    ON UPDATE CASCADE
    ON DELETE CASCADE,
```

```

CONSTRAINT ck_templates_creator_type
    CHECK (creator_type IN ( 'SYSTEM', 'USER',
        'AI' )),
CONSTRAINT ck_templates_status
    CHECK (status IN ( 'DRAFT', 'ACTIVE',
        'ARCHIVED' )),
CONSTRAINT ck_templates_name_not_blank
    CHECK (length(btrim(name)) > 0),
CONSTRAINT ck_templates_updated_after_created
    CHECK (updated_at >= created_at)
);

CREATE TABLE template_sections (
    section_id          UUID          NOT NULL
        DEFAULT gen_random_uuid(),
    template_id         UUID          NOT NULL,
    heading_title       VARCHAR(255)  NOT NULL,
    order_index         INTEGER       NOT NULL,
    prompt_instruction  TEXT          NULL,
    created_at          TIMESTAMP     NOT NULL
        DEFAULT CURRENT_TIMESTAMP,
    updated_at          TIMESTAMP     NOT NULL
        DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_template_sections PRIMARY KEY
        (section_id),

    CONSTRAINT uq_template_sections_order UNIQUE
        (template_id, order_index),
    CONSTRAINT fk_template_sections_template
        FOREIGN KEY (template_id) REFERENCES
            templates (template_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT ck_template_sections_order_index
        CHECK (order_index >= 0),
    CONSTRAINT ck_template_sections_heading_not_blank
        CHECK (length(btrim(heading_title)) > 0)
);

CREATE TABLE placeholders (

```

```

placeholder_id      UUID      NOT NULL
    DEFAULT gen_random_uuid() ,
section_id          UUID      NOT NULL,
variable_name       VARCHAR(100) NOT NULL,
expected_data_type  VARCHAR(30) NOT NULL,
description         TEXT      NULL,
is_required         BOOLEAN   NOT NULL
    DEFAULT TRUE,
created_at          TIMESTAMP  NOT NULL
    DEFAULT CURRENT_TIMESTAMP,
updated_at          TIMESTAMP  NOT NULL
    DEFAULT CURRENT_TIMESTAMP,
CONSTRAINT pk_placeholders PRIMARY KEY
    (placeholder_id),

CONSTRAINT uq_placeholders_section_variable
    UNIQUE (section_id, variable_name),
CONSTRAINT fk_placeholders_section
    FOREIGN KEY (section_id) REFERENCES
        template_sections (section_id)
    ON UPDATE CASCADE
    ON DELETE CASCADE,
CONSTRAINT ck_placeholders_expected_data_type
    CHECK (expected_data_type IN ( 'STRING',
                                    'NUMBER', 'BOOLEAN',
                                    'DATE', 'LIST',
                                    'OBJECT',
                                    'CODE' ) ),

CONSTRAINT ck_placeholders_variable_name_format
    CHECK (variable_name ~
        '^[A-Za-z_][A-Za-z0-9_]*$')
);

CREATE TABLE template_mappings (
    mapping_id      UUID      NOT NULL
        DEFAULT gen_random_uuid() ,
    placeholder_id  UUID      NOT NULL,
    code_entity_id  UUID      NOT NULL,
    extraction_rule  TEXT      NULL,

```



```

        is_active          BOOLEAN          NOT NULL
        DEFAULT TRUE,
    CONSTRAINT pk_template_mappings PRIMARY KEY
        (mapping_id),

    CONSTRAINT uq_template_mappings_pair UNIQUE
        (placeholder_id, code_entity_id),
    CONSTRAINT fk_template_mappings_placeholder
        FOREIGN KEY (placeholder_id) REFERENCES
            placeholders (placeholder_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT fk_template_mappings_code_entity
        FOREIGN KEY (code_entity_id) REFERENCES
            code_entities (code_entity_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE
    );

CREATE TABLE documents (
    document_id            UUID              NOT NULL
        DEFAULT gen_random_uuid(),
    workspace_id           UUID              NOT NULL,
    author_user_id         UUID              NOT NULL,
    title                  VARCHAR(255)      NOT NULL,
    document_type          VARCHAR(30)       NOT NULL,
    status                 VARCHAR(20)       NOT NULL
        DEFAULT 'DRAFT',
    created_at             TIMESTAMP         NOT NULL
        DEFAULT CURRENT_TIMESTAMP,
    updated_at             TIMESTAMP         NOT NULL
        DEFAULT CURRENT_TIMESTAMP,
    deleted_at             TIMESTAMP         NULL,
    CONSTRAINT pk_documents PRIMARY KEY (document_id),
    CONSTRAINT fk_documents_workspace
        FOREIGN KEY (workspace_id) REFERENCES
            workspaces (workspace_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT fk_documents_author

```

```

        FOREIGN KEY (author_user_id) REFERENCES users
            (user_id)
        ON UPDATE CASCADE
        ON DELETE RESTRICT,
    CONSTRAINT ck_documents_type
        CHECK (document_type IN ( 'SRS', 'API_DOC',
            'USER_GUIDE',
                                   'RELEASE_NOTE',
                                   'TECH_DESIGN',
                                   'OTHER' )),
    CONSTRAINT ck_documents_status
        CHECK (status IN ( 'DRAFT', 'IN_REVIEW',
            'PUBLISHED', 'ARCHIVED' )),
    CONSTRAINT ck_documents_title_not_blank
        CHECK (length(btrim(title)) > 0),
    CONSTRAINT ck_documents_deleted_after_created
        CHECK (deleted_at IS NULL OR deleted_at >=
            created_at)
);

```

CREATE UNIQUE INDEX

```

uq_documents_workspace_title_active
ON documents (workspace_id, title)
WHERE deleted_at IS NULL;

```

```

CREATE TABLE document_versions (
    version_id          UUID          NOT NULL
        DEFAULT gen_random_uuid() ,
    document_id         UUID          NOT NULL,
    version_number      VARCHAR(20)   NOT NULL,
    content             TEXT          NULL,
    change_summary      TEXT          NULL,
    status              VARCHAR(20)   NOT NULL
        DEFAULT 'DRAFT' ,
    ticket_codes        VARCHAR(500)  NULL,
    created_at          TIMESTAMP     NOT NULL
        DEFAULT CURRENT_TIMESTAMP,
    updated_at          TIMESTAMP     NOT NULL
        DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_document_versions PRIMARY KEY

```

```

        (version_id),

    CONSTRAINT uq_document_versions_number UNIQUE
        (document_id, version_number),
    CONSTRAINT fk_document_versions_document
        FOREIGN KEY (document_id) REFERENCES
            documents (document_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT ck_document_versions_status
        CHECK (status IN ('DRAFT', 'IN_REVIEW',
            'APPROVED', 'REJECTED', 'PUBLISHED')),

    CONSTRAINT ck_document_versions_number_format
        CHECK (version_number ~
            '^[vV]?[0-9]+(\.[0-9]+){0,2}$')
);

CREATE TABLE change_logs (
    log_id          UUID          NOT NULL
        DEFAULT gen_random_uuid(),
    version_id      UUID          NOT NULL,
    change_type     VARCHAR(20)   NOT NULL,
    change_description TEXT       NULL,
    diff_content    TEXT          NULL,
    created_at      TIMESTAMP     NOT NULL
        DEFAULT CURRENT_TIMESTAMP,
    updated_at      TIMESTAMP     NOT NULL
        DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_change_logs PRIMARY KEY (log_id),
    CONSTRAINT fk_change_logs_version
        FOREIGN KEY (version_id) REFERENCES
            document_versions (version_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT ck_change_logs_change_type
        CHECK (change_type IN ('CREATE', 'UPDATE',
            'DELETE', 'RESTORE', 'PUBLISH'))
);

```

```

CREATE INDEX idx_templates_workspace_id ON templates
    (workspace_id);
CREATE INDEX idx_templates_status ON templates
    (status);

CREATE INDEX idx_template_mappings_code_entity ON
    template_mappings (code_entity_id);
CREATE INDEX idx_template_mappings_active ON
    template_mappings (placeholder_id)
    WHERE is_active = TRUE;

CREATE INDEX idx_documents_workspace_id ON documents
    (workspace_id);
CREATE INDEX idx_documents_author ON documents
    (author_user_id);
CREATE INDEX idx_documents_status ON documents
    (status);
CREATE INDEX idx_documents_created_at ON documents
    (created_at DESC);

CREATE INDEX idx_document_versions_status ON
    document_versions (status);
CREATE INDEX idx_document_versions_created_at ON
    document_versions (document_id, created_at DESC);

CREATE INDEX idx_change_logs_version_id ON
    change_logs (version_id);
CREATE INDEX idx_change_logs_created_at ON
    change_logs (created_at DESC);

CREATE OR REPLACE FUNCTION set_updated_at()
RETURNS TRIGGER AS $$
BEGIN
    NEW.updated_at := CURRENT_TIMESTAMP;
    RETURN NEW;
END;

CREATE TRIGGER trg_templates_updated_at
    BEFORE UPDATE ON templates

```

```

        FOR EACH ROW EXECUTE FUNCTION set_updated_at();
CREATE TRIGGER trg_template_sections_updated_at
    BEFORE UPDATE ON template_sections
        FOR EACH ROW EXECUTE FUNCTION set_updated_at();
CREATE TRIGGER trg_placeholders_updated_at
    BEFORE UPDATE ON placeholders
        FOR EACH ROW EXECUTE FUNCTION set_updated_at();
CREATE TRIGGER trg_documents_updated_at
    BEFORE UPDATE ON documents
        FOR EACH ROW EXECUTE FUNCTION set_updated_at();
CREATE TRIGGER trg_document_versions_updated_at
    BEFORE UPDATE ON document_versions
        FOR EACH ROW EXECUTE FUNCTION set_updated_at();
CREATE TRIGGER trg_change_logs_updated_at
    BEFORE UPDATE ON change_logs
        FOR EACH ROW EXECUTE FUNCTION set_updated_at();
GO

```


Chương 6

Extensions

Phần III

Conclusion

Chương 7

Summary

Chương 8

Future research

Phần IV

Technical appendix

Phụ lục A

Generic technical topic

Phụ lục B

Another technical topic

Phụ lục C

Last technical topic

This is a last appendix with a bit of text.

C.1 A first section

Sed vestibulum ex a tristique lacinia. Integer interdum magna vel magna rutrum fermentum. Maecenas sed mi in nunc convallis rutrum. Vivamus dapibus bibendum est, ac tincidunt ipsum fermentum id. Nunc id est turpis. Suspendisse potenti. Sed ac laoreet nulla, eu sollicitudin libero. Suspendisse tempus orci nec mauris volutpat eleifend. Integer nec tristique libero. Sed vehicula ipsum sit amet magna accumsan, nec pulvinar nisl consequat. In ac hendrerit turpis. Sed aliquet luctus mauris, vitae congue turpis. Nullam blandit lacus vel interdum eleifend.

C.2 A second section

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$$S^*(z) = \frac{\alpha}{1 - \gamma(1 - \alpha)} > \int \frac{\alpha(i)}{\gamma(i)} di. \quad (\text{C.1})$$

Nam pretium mauris eros, nec sollicitudin risus tincidunt a. Vivamus in augue vitae ligula scelerisque dapibus. Integer eget metus aliquet, efficitur

nibh eget, pharetra eros. Suspendisse luctus interdum ex id suscipit. Donec quis augue mauris. Nulla ut erat eget nisl hendrerit malesuada. Vivamus eu tortor sit amet sem fringilla eleifend. In hac habitasse platea dictumst. Suspendisse potenti. Ut eget libero sed orci tempor ullamcorper:

$$\mathcal{W}^2(\zeta) = \left[\frac{\beta}{1 - \nu(1 - \xi)} \right] \pm \Lambda \quad (\text{C.2})$$

$$\mathcal{L}^2(\zeta) = \left(\frac{\beta}{1 - \nu(1 - \xi)} \right) \quad (\text{C.3})$$

$$\overline{a - b} = \int_0^{+\infty} \frac{\beta}{1 - \nu(x)(1 - \xi)} dx \in \mathbb{R}. \quad (\text{C.4})$$

C.3 A third section

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Tài liệu tham khảo

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Nemo enim ipsam voluptatem, quia voluptas sit, aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos, qui ratione voluptatem sequi nesciunt, neque porro quisquam est, qui dolorem ipsum, quia dolor sit amet consectetur adipiscing velit, sed quia non numquam eius modi tempora incidunt, ut labore et dolore magnam aliquam quaerat voluptatem. At vero eos et accusamus et iusto odio dignissimos ducimus, qui blanditiis praesentium voluptatum deleniti atque corrupti, quos dolores et quas molestias excepturi sint, obcaecati cupiditate non provident, similique sunt in culpa, qui officia deserunt mollitia animi, id est laborum et dolorum fuga.